Plan or Learn? Dynamic Pricing of Airline Tickets under Model Error

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Code: `https://github.com/kadridiaa/airline-dynamic-pricing-RL`

1 Introduction

An airline sells the seats of a flight over 30 days and sets a price every day. With a known demand model, Dynamic Programming (DP) computes the optimal policy; in practice the model is estimated, hence wrong, and planning with a wrong model is sub-optimal [1, §8.3]. Q-learning needs no model but many episodes. **Research question:** from which model error does learning without a model outperform planning with a wrong one, for a given budget of episodes?

2 Problem formulation

MDP. State $s = (c, t)$: $c \in \{0..10\}$ seats left, $t \in \{0..30\}$ days to departure (keeping t in the state makes it Markov). Actions: six prices $p \in \{50, 80, \ldots, 200\}$€. Selling k seats gives $r = p\,k$ and $(c, t) \to (c - k, t - 1)$; episodes end at $t = 0$ or $c = 0$; $\gamma = 1$. This gives 341 states and 2 046 Q-values.

Demand. Leisure customers arrive at rate $\lambda_L = 0.8$/day (0.3 when $t \le 10$), business customers at $\lambda_B = 0.1$ (1.0 when $t \le 10$). Willingness to pay (WTP) is exponential with mean $m_L = 80$€, $m_B = 180$€, so a customer buys with probability $e^{-p/m}$. By Poisson thinning, demand is $D \sim \text{Poisson}(\mu)$, $\mu = \lambda_L e^{-p/m_L} + \lambda_B e^{-p/m_B}$, and sales $k = \min(D, c)$. We use $C = 10$ seats: with 20, the plane almost never sells out and a myopic policy loses only 0.1 %, against 18.9 % with 10 (load factor 77 %).

3 Methods

DP. Bellman optimality, $V^*(c, t) = \max_a \sum_k P(k \mid c, t, a)\,[p_a k + \gamma V^*(c - k, t - 1)]$, with $V^*(c, 0) = V^*(0, t) = 0$. As $V^*(\cdot, t)$ only depends on $V^*(\cdot, t - 1)$, one backward sweep is exact.

Model error. The DP plans with both WTP means multiplied by $(1 + x)$, $x \in [-50, +50]$ % (an error on λ costs at most 5 %).

Q-learning, SARSA. Tabular, $Q = 0$, ε-greedy with ε decaying exponentially from 1 to 0.01 over the episode budget; $\alpha_n = 0.5/(1 + 40\,n/N)$ at episode n of a budget of N (a constant α does not converge in this stochastic environment: Q-values keep oscillating and the greedy action flips). These values were selected among 9 configurations by the best worst case over seeds. Both methods only interact with the simulator, never with the demand model.

Baselines. Best fixed price; best linearly rising price.

Evaluation. All policies are scored on the *true* model by exact policy evaluation (cross-checked with 10 000 simulated flights), as a percentage of the optimal revenue; learning methods use 10 seeds and are reported with 95 % confidence intervals of the mean (Student). Checks: the DP value lies in the 95 % confidence interval of its simulated revenue; no wrong-model DP or baseline beats it. These checks and unit tests of every component run automatically (56 `pytest` tests), and each figure is reproduced exactly by one script.

4 Results

Policy	Revenue	% opt.	Load
DP (optimum)	1 193.5 €	100.0	77 %
Rising price 140→170 €	1 166.1 €	97.7	74 %
Fixed price 170 €	1 146.7 €	96.1	68 %
Q-learning (30k)	1 141.5 €	95.6 ± 0.3	84 %
SARSA (30k)	1 132.2 €	94.9 ± 0.3	85 %
Q-learning (100k)	1 150.3 €	96.4 ± 0.5	86 %
SARSA (100k)	1 150.2 €	96.4 ± 0.2	86 %

Table 1: Expected revenue per flight on the true model. Learning methods: mean over 10 seeds ± 95 % CI (episodes in brackets).

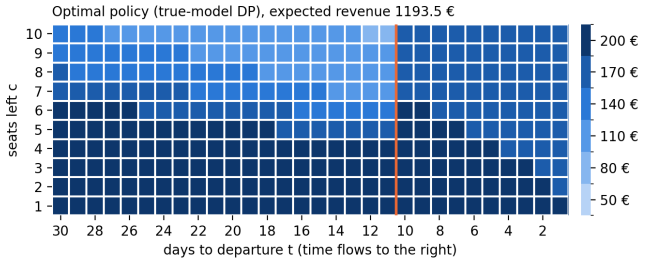


Figure 1: Optimal DP policy. Business customers arrive from $t = 10$ (orange line).

Optimal policy (Fig. 1). The price never increases with the seats left, as predicted by [2]; it jumps when business customers arrive, and drops to 80 € at $t = 11$–12 to sell remaining seats to leisure customers. A rising price already reaches 97.7 % (Table 1), consistent with the near-optimality of simple pricing heuristics [2].

E2: tipping point between planning and learning (Fig. 2). The DP stays above 97 % for $|x| \le 20$ %. Errors are asymmetric: underestimating the WTP ($x = -50$ %) drops DP to 75.8 % because seats sell out too cheaply, against 92.9 % at +50 %. Model-free RL therefore beats the wrong DP, significantly at the 95 % level, with 1 000 episodes at $x = -50$ % and 10 000 at −40 % and −30 %. When the WTP is overestimated, the DP is more resilient: RL only wins at +40 % (30 000 episodes) and +50 % (10 000). For $-20\,\% \le x \le +30\,\%$, the DP wins or ties at every budget. Tuned on the same wrong model, the rising-price rule is below the wrong DP at every x (e.g. 66.0 % at $x = -50$ %): its 97.7 % comes from knowing the true model. With 66 tests at the 95 % level, a few isolated cells may be significant by chance; our conclusions rest on the large contiguous regions, not on single cells.

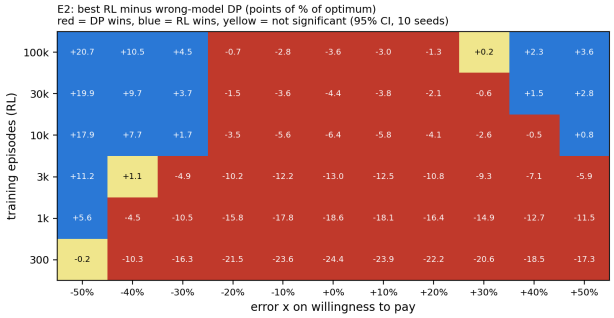


Figure 2: E2: winning method across model error and episode budget (blue = RL wins, red = DP wins, yellow = not significant at 95 %).

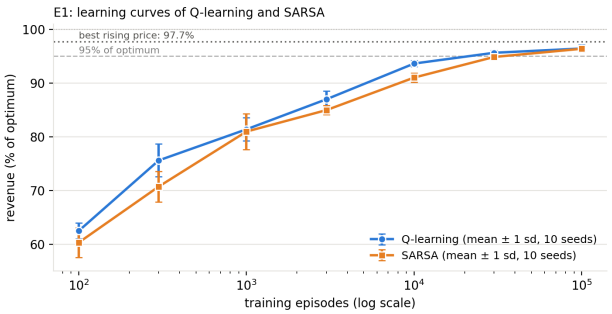


Figure 3: E1: learning curves of Q-learning and SARSA (mean ± 1 sd over 10 seeds, log scale).

E1 & E3: learning budget and policy heatmaps (Figs. 3 and 4). Q-learning reaches 95 % of the optimum from 30 000 episodes (95.6 %, CI [95.3, 96.0]), and both methods plateau at 96.4 % at 100 000 episodes, significantly below the rising-price heuristic (97.7 %; upper CI bound 97.0 %). Off-policy Q-learning converges faster at intermediate budgets (93.6 % vs 91.0 % at 10k), while SARSA varies less across seeds at convergence (standard deviation 0.3 vs 0.7 points at 100k). Learned policies fill the plane more than the DP (84–86 % vs 77 %, Table 1): they sell too many seats too cheaply. Heatmaps show that RL captures the structure of the optimal policy; 21 of the 300 states (few seats left very early) are never learned because they are almost never reached, and rarely visited states remain noisy at little cost in revenue.

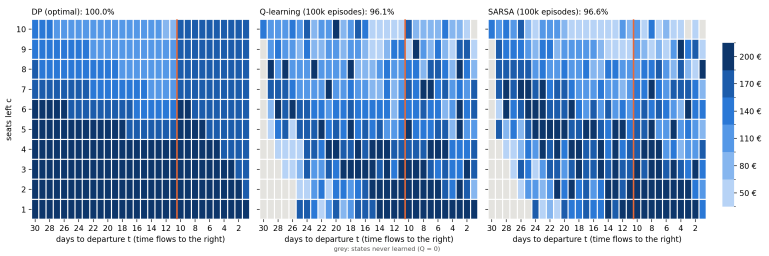


Figure 4: E3: policies of the DP, Q-learning and SARSA (100k episodes); grey states are never visited.

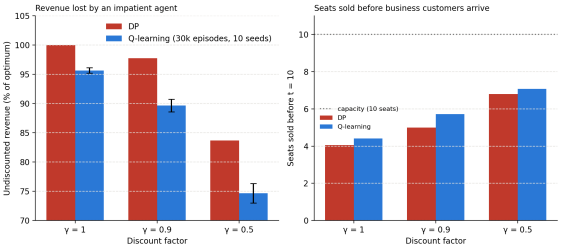


Figure 5: E4: revenue on the true model (left) and seats sold before business customers arrive (right).

E4: discount factor (Fig. 5). A DP optimising a discounted criterion sells seats too early: with $\gamma = 0.5$ it sells 6.8 seats before business customers arrive (4.1 with $\gamma = 1$) and earns 83.7 % of the optimum (97.8 % with $\gamma = 0.9$). Q-learning (30 000 episodes, 10 seeds) shows the same myopia: 95.6 % ($\gamma = 1$), 89.7 % ($\gamma = 0.9$) and 74.7 % ($\gamma = 0.5$), selling 7.1 seats before business customers arrive.

5 Conclusion

Learning outperforms planning only for large model errors: an under-estimation of the WTP by 30 % or more (with 10^3 to 10^4 flights), or an over-estimation by 40 % or more (10^4 to $3 \cdot 10^4$ flights). For moderate errors ($-20\,\% \le x \le +30\,\%$) or scarce data, planning with an imperfect model beats model-free learning. Even with 10^5 flights, tabular RL stays below a simple rising-price rule tuned on the true model: the value of planning (or of a good model) lies in the last few percent.

Limitations and perspective. One episode is one flight: 30 000 episodes would be about 82 years of daily flights, so in practice RL itself would need a simulator, i.e. a model; this strengthens the case for planning. The RL hyper-parameters were tuned by scoring on the true model, which favours RL; tuning on independent seeds or a held-out model would only make RL weaker. Demand is synthetic, only the WTP is perturbed, and methods are tabular: realistic aircraft would require value function approximation, and learning the model from data (model-based RL, Dyna [1, ch. 8]) is the natural next step between our two extremes.

References

[1] R. S. Sutton, A. G. Barto. *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

[2] G. Gallego, G. van Ryzin. Optimal dynamic pricing of inventories with stochastic demand over finite horizons. *Management Science* 40(8), 1994.